5. Bar Plot of Trading Volume of Alphabet Inc.

Aim

To write a Pandas program to create a **bar plot of the trading volume of Alphabet Inc. stock between two specific dates**.

Algorithm

1. Import Pandas and Matplotlib.
2. Read the Alphabet stock Excel dataset.
3. Convert the Date column into datetime format.
4. Specify the starting and ending dates.
5. Filter the records between the specified dates.
6. Create a bar plot using Date and Volume.
7. Add title and axis labels.
8. Display the plot.
9. Stop.

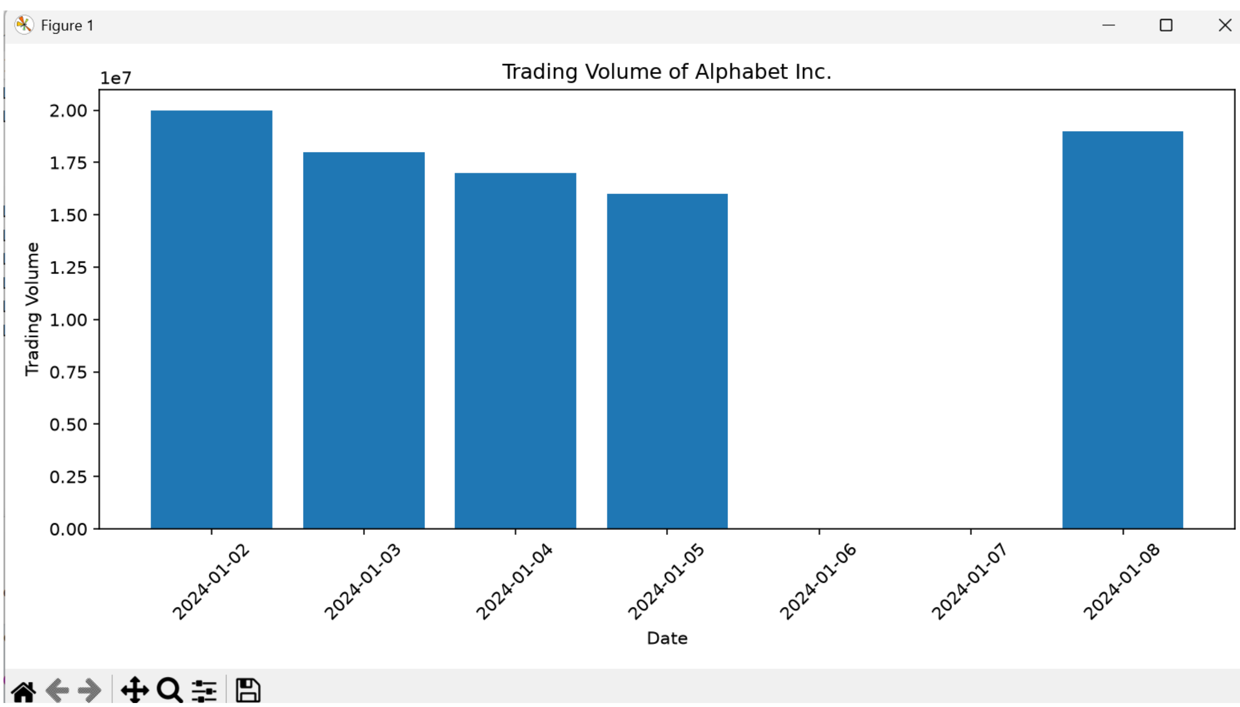
Dataset

Date	Open	High	Low	Close	Volume
2024-01-02	138.55	140.62	138.41	139.69	20000000
2024-01-03	138.60	139.94	137.74	138.10	18000000
2024-01-04	138.12	139.50	137.78	138.04	17000000
2024-01-05	138.35	139.90	138.20	138.87	16000000
2024-01-08	139.00	141.20	138.80	140.10	19000000

```python
import pandas as pd
import matplotlib.pyplot as plt
# Read Excel dataset
stock = pd.read_excel(r"C:\Users\DELL\Desktop\dsa0404 4_files\alphabet.xlsx")
# Convert Date column to datetime
stock["Date"] = pd.to_datetime(stock["Date"])
# Specify date range
start_date = "2024-01-01"
end_date = "2024-12-31"
# Filter data
filtered_stock = stock[
    (stock["Date"] >= start_date) &
    (stock["Date"] <= end_date)
]
# Create bar plot
plt.figure(figsize=(10, 5))
plt.bar(
    filtered_stock["Date"],
```

```python
plt.figure(figsize=(10, 5))
plt.bar(
    filtered_stock["Date"],
    filtered_stock["Volume"]
)
plt.title("Trading Volume of Alphabet Inc.")
plt.xlabel("Date")
plt.ylabel("Trading Volume")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

OUTPUT:



Result

Thus, the Pandas program was successfully executed to create a bar plot showing the trading volume of Alphabet Inc. between two specified dates.

6. Scatter Plot of Trading Volume / Stock Prices of Alphabet Inc.

Aim

To write a Pandas program to create a **scatter plot showing the relationship between trading volume and stock price of Alphabet Inc. between two specific dates**.

Algorithm

1. Import Pandas and Matplotlib.
2. Read the Alphabet stock Excel dataset.
3. Convert the Date column to datetime format.
4. Specify the starting and ending dates.
5. Filter the stock records within the specified dates.
6. Use Volume on the X-axis.
7. Use Close price on the Y-axis.
8. Create a scatter plot.
9. Add title and axis labels.
10. Display the plot.
11. Stop.
12. **Dataset**

Date	Open	High	Low	Close	Volume
2024-01-02	138.55	140.62	138.41	139.69	20000000

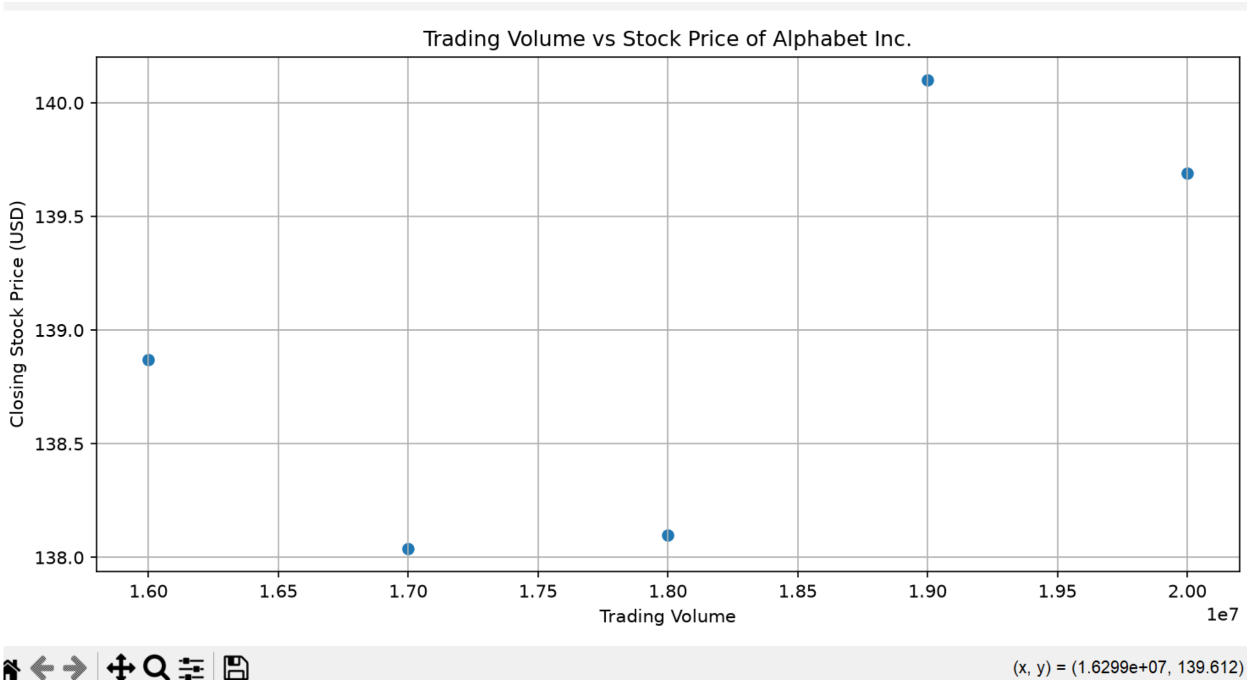
Date	Open	High	Low	Close	Volume
2024-01-03	138.60	139.94	137.74	138.10	18000000
2024-01-04	138.12	139.50	137.78	138.04	17000000
2024-01-05	138.35	139.90	138.20	138.87	16000000
2024-01-08	139.00	141.20	138.80	140.10	19000000

```python
import pandas as pd
import matplotlib.pyplot as plt
# Read Excel dataset
stock = pd.read_excel( r"C:\Users\DELL\Desktop\dsa0404 4_files\alphabet.xlsx")
# Convert Date column to datetime
stock["Date"] = pd.to_datetime(stock["Date"])
# Specify date range
start_date = "2024-01-01"
end_date = "2024-12-31"
# Filter data
filtered_stock = stock[
    (stock["Date"] >= start_date) &
    (stock["Date"] <= end_date)
]
# Create scatter plot
plt.figure(figsize=(10, 5))
plt.scatter(
    filtered_stock["Volume"],
```

filtered_stock["Volume"],

```python
    filtered_stock["Close"]
)
plt.title("Trading Volume vs Stock Price of Alphabet Inc.")
plt.xlabel("Trading Volume")
plt.ylabel("Closing Stock Price (USD)")
plt.grid(True)
plt.tight_layout()
plt.show()
```

OUTPUT:



Result

Thus, the Pandas program was successfully executed to create a scatter plot showing the relationship between trading volume and the closing stock price of Alphabet Inc. between two specified dates